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Implementing Service-Oriented Architecture in Organizations

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Services and the general chair of the IEEE Services Computing Conference 2006 and IEEE International Conference on Web Services 2008.

ABSTRACT: Service-oriented architecture (SOA) has been promoted as a technology that can enhance information systems agility, interoperability between applications, deployment flexibility, and reusability. As with any new information technology (IT), the decision to adopt SOA cannot be taken lightly, given the nontrivial investment in economic and personnel resources. The complexity associated with industry-wide diffusion, coupled with organization, industry, and environment factors, contributes to a lack of a clear strategy for assessing the business value that SOA provides an organization. This research attempts to shed light on this process of value creation for an organization, using a system dynamics approach. A detailed model of the industry diffusion coupled with organization adoption is presented. After suitable calibration and validation, a series of simulations using the model evaluate the efficacy of SOA under a variety of diverse conditions. The results of the simulations indicate clear benefits of SOA over monolithic ITs when employed under appropriate conditions. Situations where SOA fails to live up to expectations are also identified. The model and accompanying simulations can serve as a practical decision support tool for an organization to help make the strategic decisions of adopting and implementing SOA.

KEY WORDS AND PHRASES: adoption, design science, service-oriented architecture, service science, services, system dynamics.

ORGANIZATIONS ARE INCREASINGLY LOOKING TO information technology (IT) as a vehicle to improve performance and generate sustained competitive advantage [46]. In an increasingly dynamic and global business environment, IT forms a critical component in enhancing business agility, as organizations seek to become more flexible and responsive to changing business conditions. *Agility* is defined as the ability to swiftly reconfigure operations, process, and business relationships and is differentiated from flexibility (which is usually aimed at the level of a system component, whereas agility assumes multiple system components [64]). However, with the traditional monolithic architecture of IT applications, it is quite difficult, expensive, and time consuming to make changes to the applications in response to changing business conditions. The ability to quickly make changes to IT applications in response to changing business conditions is referred to as *information systems (IS) agility*.

Service-oriented architecture (SOA) has been advocated as an approach for building systems that enhances IT's ability to efficiently and effectively respond to the rapidly changing business environment and enables organizations to respond to these changes in a timely manner [11, 30]. It represents a core technology in the increasingly important discipline of service science [5]. One of the major objectives of SOA is to improve IS agility. Many definitions and characterizations of SOA appear in the literature emphasizing the underlying technology [6], operating paradigms [44], effects on the organization, and the like. Some of the definitions focus on the techni-

cal aspects of the architecture, whereas others incorporate the business perspective. From a technical viewpoint, SOA is an application architecture wherein functions or services are defined using a description language with callable interfaces to support business processes [11]. This research employs the definition provided by Arsanjani et al., who adopt the view that

SOA is the architectural style that supports loosely coupled services to enable business flexibility in an interoperable, technology-agnostic manner. SOA consists of a composite set of business-aligned services that support a flexible and dynamically re-configurable end-to-end business processes realization using interface-based service descriptions. [3].

Implicit in this definition is the objective of employing SOA to enhance IS and business agility and improve IT–business alignment in a rapidly changing business environment. One major difference between SOA and other ITs is its emphasis on enhancing IS agility, which enhances business agility. This is manifest as close alignment between IT and business, providing an organization the opportunity to quickly reconfigure IS applications in response to changing business needs.

SOA provides methods for systems development and integration where systems group functionality around business processes and package these as an interoperable service [44]. These services are typically made available for reuse within an organization or may be commercially available. Thus, SOA separates functions into distinct units, or services, which developers make accessible over a network in order that users can combine and reuse them in the production of business applications [7]. These services communicate with each other by passing data from one service to another, or by coordinating an activity between two or more services. SOA relies on service orientation as its fundamental design principle. In an SOA environment, users can access independent services without knowledge of their underlying platform implementation. SOA also describes IT infrastructure which allows different applications to exchange data with one another as they participate in business processes.

SOA is often considered a means for organizations to boost return on IT investment through reduced costs, elimination of redundancy, and improved productivity, among other [8]. Many organizations have invested in SOA expecting to reap the claimed benefit [10, 40]. Despite claims that the adoption of SOA will entail improved business value, conclusive proof is hard to come by. Many firms fail to realize immediate value for SOA as they trade off long-term benefits versus short-term local needs of project management [30]. Reported SOA success stories among peers and competitors are frequent justifications for whether adoption of the technology is a good fit for an organization. Although SOA has been perceived as a means of achieving IT integration and improving productivity, its ability to improve business agility [64] should not be understated. SOA provides an information architecture that permits quicker and more responsive change to applications through the selection and integration of appropriate services. This nimble response is not usually afforded by legacy applications. A quicker response to changes in business processes gives an organization an advantage over its competitors. In addition, the use of available services also provides an organization

with benefits of higher quality and reduced maintenance through reuse. Furthermore, SOA can serve as a vehicle to align a firm's IT strategy with its business strategy and thus can enhance a firm's agility.

The adoption of SOA is different from other ITs because it is an architectural style that affects the entire IS organization and how it responds to changes in business, and consequently places a spotlight on IT–business alignment. In contrast, the adoption of most other ITs has a more limited impact. The business value derived from SOA by an organization is determined by several factors, including the competitiveness of the firm's industry, the appropriate level of business agility, implementation effectiveness, system security and performance requirements, and organizational innovativeness, among others. Further, the complex and dynamic nature of the factors affecting the SOA adoption and implementation process makes it difficult to assess the effect of strategic decisions surrounding SOA adoption and implementation.

This research seeks to examine the impact of adopting SOA on a firm's agility, particularly in terms of response to changes in underlying business processes as compared to traditional architected applications. It is convenient to study the response at an individual organization level, but it must be borne in mind that the adoption of SOA by one organization is not an independent decision, and is based in large part on the extent of diffusion within the industry. Current technology adoption and diffusion models [6, 31, 49] tend to examine the innovation diffusion phenomenon separately from the impact of the firm's adoption of the innovation. This research attempts to integrate the two phenomena by creating a model that encompasses industry-level SOA diffusion as well as firm-level impact on agility. Given the prevailing organizational, environmental, and contextual factors, firms need to evaluate alternative courses of action to arrive at the most appropriate strategy for adopting and implementing SOA in their organization. The inability to effectively assess the consequences of strategic decisions *a priori* leaves managers to speculate if the decisions made were correct. A model that captures the complexities, dynamics, and a range of decision options while providing managers with the ability to explore various decision options and examine outcomes of these decisions would be invaluable to managers. This paper takes a design science approach [32, 42, 45] to developing a modeling suite that enables managers to accomplish this.

Theoretical Background

THE ADOPTION AND DIFFUSION OF INNOVATIONS BY ORGANIZATIONS is a discipline with a rich tradition [14, 16, 37, 58]. Innovation adoption research encompasses the generation, development, and implementation of new ideas or behaviors [16] and is based on the premise that it is a decision to make full use of an innovation as the best course of action available [49]. Although it is an organizational phenomenon, many researchers model adoption as an individual decision process, leading to criticisms that technology adoption studies have failed to grasp a complete picture due to the reliance on an individual adoption decision perspective [22]. In an effort to reconcile these differences, two different approaches were employed. One approach employs an organizational

perspective, incorporating innovation characteristics [56], organizational size, slack, and professional orientation of the IS unit [28], among others. Swanson [55] categorized innovations into three types—functional IS core, administrative IS core, and core business technology. Grover et al. [28] extend the organizational perspective and empirically show that organizational size, slack, and professional orientation of the IS unit affect the adoption and diffusion phenomenon. An alternative approach involves the extension of the individual decision to incorporate organizational constructs such as the characteristics of user community or organizational environment, leading to the integrated acceptance model known as unified theory of acceptance and use of technology model [61].

Moving beyond adoption, diffusion represents the process wherein members of a social system communicate and over time all or most members adopt the innovation [59]. Rogers [49] defines diffusion as a process in which an innovation is communicated through specific channels. In a similar manner, diffusion has been studied at two distinct levels—that is, organizational and industrial levels. Studies on organizational-level diffusion examine the organizational adoption decision and its success, whereas research on industry-level diffusion study the success of technology within an industry. In both streams, institutional theory [17] forms a basis for examining the issues, suggesting that coercive, mimetic, and normative pressures are core factors affecting innovation. The notion of normative pressure is rooted on network theory, which assumes the existence of relatively homogeneous organizations within an industry. Roberts and Greenwood [47] counter that institutional theories explain only a part of the diffusion phenomenon, that is, nonchoice behavior, which Swanson and Ramiller characterize as “mindlessness” [56]. Instead, they suggest that the adoption decision is based on an efficiency-seeking attitude using transaction cost theory. Building on this, Westphal et al. [63] proposed interweaving transaction cost theory and institutional theory to explain adoption decisions, suggesting that efficiency is a major driver for the early adopters while late followers are influenced more by legitimacy.

Like many new ITs, SOA adoption has been characterized by anecdotal successes and failures. Most of the prescriptions for success and documentations of failure focus on the characteristics of the innovation. These include complexity of the technology, instability of open standards for SOA, shortcuts made in speed-to-market decisions, performance issues, and anticipated changes in SOA application, among others [11, 12, 30, 60]. It is common for the first release of an SOA application to take additional time because adherence to SOA design principles often leads to longer design time without yielding immediate benefits [30]. In addition to the learning curve associated with any new IT, firms may not be able to discover promised long-term SOA benefits in short-lifespan applications because the major portion of return on investment is realized as applications evolve over time. Moreover, a variety of misconceptions on SOA lead firms to greatly underestimate the effort required to successfully implement SOA and make them more susceptible to vendor advertising and hype [39]. Lewis et al. reiterate that “SOA is not an architecture but an architectural pattern from which an infinite number of architectures can be derived—both good and bad” [39, p. 123]. This ambiguity and complexity in evaluating SOA for a specific firm makes understanding

the SOA adoption and diffusion phenomenon different from other technologies. It has been suggested that SOA is a valuable approach when an application is expected to undergo significant change in platform requirement, load requirement, integration requirement, and scalability, among others. If the application in question is isolated from many of the other systems in the business, or if, over the life of the application, the requirements and demands of the application are relatively stagnant, SOA may not be the most appropriate approach. These diverse approaches to SOA adoption provide different facets that have an impact on the eventual success of SOA diffusion.

The SOA adoption process is characterized by a high adoption barrier. It is not an easy decision to adopt SOA, because most firms cannot afford giving up short-term benefits when project management goals and the necessity to adhere to SOA design principles conflict with each other [30]. Small businesses have largely been left out of the SOA transformation and very few small businesses believe that SOA can yield tangible benefits for them [12]. This low imitability promotes rarity, leading to potentially high value of SOA innovation for its adopters [62]. In a rapidly changing environment, however, the rarity stemming from low imitability alone cannot sustain competitive advantage of innovation in the long run [21] and dynamic capabilities need to be analyzed. While prescriptive checklists may play a role in the context of one organization's adoption decision, a more holistic approach is needed to better understand the SOA adoption and diffusion phenomenon due to the complexity and dynamism involved in the focal problem domain. Some studies examine this at an industry level [6], whereas others study the process from an individual firm's perspective [31]. However, attempts to examine a firm's adoption decision in the context of diffusion within the industry are lacking. We conducted a pilot study to incorporate the organization's decision as well as industry-level diffusion [13]. While the pilot study provides reasonable results and considerable insight into the effect over time, it represents a narrow portion of possible scenarios that an organization may face. The current study significantly extends this model with additional constructs that are specific to SOA with an extensive scenario analysis through simulation.

This paper develops a modeling suite that enables managers to calibrate and design decision support tools for their own organization when they are contemplating an SOA adoption decision, using the design science research method [32, 42, 45]. Unlike standard social science research that develops a research model with hypotheses that can be tested statistically through controlled manipulation and empirical validation, the design science methodology seeks to create IT artifacts that are intended to solve specific organizational problems and provide rigorous evaluation of these artifacts based on utility rather than an empirical test of theories [32]. The design science research methodology encompasses successive steps of problem identification, definition of objectives for a solution, design or development of a suitable IT artifact, and demonstration of the proof of concept, evaluation, and communication.

The design science paradigm has its roots in engineering and is well recognized in the IS discipline [26, 53]. Although the design science approach has been slow to diffuse into the mainstream of IS research [45], it has been argued that the relevance of IS research is directly related to its applicability in design [9]. The design science

research method has the potential to provide significant insight into SOA adoption and diffusion because existing theory is insufficient to design useful artifacts in the focal problem domain [32]. This lack of a sufficient knowledge base makes the design of creative artifacts in the SOA adoption and diffusion problem challenging but vital: “The resultant IT artifacts extend the boundaries of human problem solving and organizational capabilities by providing intellectual as well as computational tools. Theories regarding their application and impact will follow their development and use” [32, p. 76].

The organizational problem being studied in this research is the impact of SOA adoption on a firm’s IS agility that enhances the IS–business alignment, given that this adoption occurs within a larger industry-level SOA diffusion context. Given that post hoc analysis of individual adoption decisions are likely to have little generalizability, and are rarely indicative of possible effects of alternative decisions, a predictive capability was sought in whatever solution strategy was chosen. The solution approach must be robust, reliable, and true to the real-world phenomenon. Because SOA adoption and diffusion is a dynamic phenomenon, the IT artifact proposed in this research is a modeling suite that allows decision makers to evaluate the impact of alternative decisions regarding SOA adoption under different environmental and industry conditions. The basis for the core model is presented in the next section, and the detailed structure is discussed in the fourth section. The model is based on constructs and relationships grounded in the theoretical framework discussed. It incorporates an organizational perspective and the individual adoption decision perspective [4, 28, 56, 61]. Institutional theory [17] is utilized to address the effect of diffusion maturity [47, 56, 63]. It includes factors present in any technology innovation [49] as well as SOA-specific factors [11, 30, 60]. Detailed theoretical underpinnings (e.g., [6, 11, 12, 14, 27, 31, 33, 36, 38, 39, 43, 48, 49, 50, 57]) of model constructs and relationships are also addressed when the model is discussed in detail in the fourth section. Execution of the models under a variety of conditions serves as the basis for demonstration of the IT artifact utility. The evaluation of the artifact is performed using a rigorous approach consistent with the modeling methodology adopted in the creation of the artifact.

Modeling Methodology

SEVERAL MODELING TECHNIQUES ARE AVAILABLE FOR STUDYING dynamic relationships between organizational constructs addressing SOA adoption and implementation decisions. This study posits that relationships between organizational, environmental, and contextual factors cannot be modeled in simple terms, and that scrutiny of performance over time is necessary to understand the underlying phenomena. The complex relationship between organizational, environmental, and contextual factors will involve nonlinearity, discontinuity, and stochastic changes in the model. In addition, preference and utility trade-offs for a manager’s risk propensity will also affect the adoption decision. As a consequence, deterministic solution strategies that seek to determine a unique solution are of little value in this case. Mathematical programming approaches have been employed in many design problems. The decision to adopt SOA

is complex. Coupled with the dynamic factors and individual differences that managers bring to the decision, this rules out mathematical programming as an appropriate solution. Several individual and collective decision-making models have been proposed, including decision making under conditions of certainty, uncertainty, and risk, among others. To varying degrees, these models assume knowledge of the alternatives at hand, their relative impact toward the decision maker's objectives, and the underlying utility and preference functions of the decision maker, all of which are hard to come by in the case of an SOA adoption decision. More importantly, these models also assume a point-in-time decision with an immediate evaluation of the outcome of the decision. This is hardly the case for adoption of SOA, wherein the benefits are expected to be reaped over a period of time in which the business environment and organizational conditions are changing dynamically.

Simulation represents a viable option under the circumstances. It provides the ability to study the adoption and diffusion as a dynamic phenomenon, as well as evaluate the decision over time under a wide variety of circumstances. Although simulation has been widely used in the manufacturing context for decades, it is only recently that it has found broader use in social science research [25, 41]. Simulation provides an advantage over traditional empirical approaches in that it permits the inclusion of all relevant variables, even if data are not readily available, and allows the researcher to model data and relationships without sacrificing accuracy for elegance [54]. It is important to acknowledge that simulation does not provide a definitive solution to the problem being studied. Rather, it serves as a mechanism to evaluate if the decision made supports the objectives of the decision maker. Thus, for example, it cannot recommend an optimal period when to adopt SOA. Rather, it can explicate to the decision maker the relative pros and cons of early versus late adoption of SOA. Simulation requires careful attention to detail during model construction and calibration; otherwise the model will not reflect reality, and decision makers will be relying on inaccurate portrayals to evaluate decisions. Rigorous testing of the model and comparison with proven real-world results are needed to offset this concern. Last, a simulation model must be robust so that it permits the decision maker to evaluate the entire decision space effectively and without brittleness of results. The need for robustness and accuracy with respect to real-world conditions layers additional complexity on the model, but gives the decision maker the ability to evaluate the decision space more systematically, and affords greater confidence in the decisions.

Several different simulation strategies are available, including discrete event versus continuous simulations and stochastic versus deterministic simulations. Discrete event simulations typically address requests for service from resource providers and are driven by probability distributions for arrival rates and service times. Continuous simulations are favored for engineering applications, where differential equations drive the relationship between constructs over time. Stochastic simulations rely on random numbers to model events and may be sensitive to the random number seeds chosen in the simulation. Deterministic simulations assume well-defined inputs and relationships between constructs. For the problem at hand—namely, the impact of SOA adoption on a firm's IT agility—it is important to understand how a construct behaves over time

and how it is related to other constructs. This deterministic though continuous simulation can be addressed through system dynamics. System dynamics [23] was designed specifically to address these issues, stressing the development of useful models. It has proved helpful for testing macro-sociological theories [34], including environmental policy, corporate strategy, health care, operations management, change management [15], and strategic management [51], among others. Although system dynamics has not been extensively used in the IS field, a strong case for its application to IS research has been made [52], particularly in cases involving novel representations of real-world situations. Some notable examples of its use in IS include its application to business planning for network services [18], diffusion of the Internet in developing countries [19], offshore outsourcing [20], and software project management [1, 2]. A primary advantage of system dynamics is that it facilitates the representation of both quantifiable and hard-to-measure variables [18]. As with any modeling approach, concerns exist about the potential use of simplifying assumptions, possible inaccuracies in modeling vis-à-vis the real world, and a lack of comprehensiveness in the final model. These can be alleviated with validation of the model through multiple strategies.

System dynamics has its roots in systems of difference and differential equations [23]. Models in system dynamics are comprised of stocks, flows, converters, and connectors as shown in Table 1. Stocks and flows are used to model organizational processes, wherein stocks represent constructs that can accumulate or deplete over time. Typically these are used to model resources—tangible or not. Flows cause a decrease or an increase in stock levels, representing resource utilization and replenishment over time. Stocks and flows are governed by conserving laws, which ensure that future stock levels are based on current levels, moderated by any flows. In contrast to stocks, converters hold inputs, outputs, and intermediate values to perform computation without accumulating values. Converters are linked to other constructs using connectors. Unlike flows, connectors represent information flows, and involve no conservation. Positive (self-reinforcing) and negative (self-correcting) feedback loops play an important role in determining dynamic behavior because most complex behaviors usually arise from feedback among the system components, not from the complexity of the components [54].

When using system dynamics, the model construction and validation procedures are closely interwoven. Assembling the model requires the initial identification of stocks and converters that form the basis for the model. A series of textual analysis of published academic and professional literature identifies stocks, converters, flows, and connectors. Creating a model based on firm-specific data is intentionally avoided so that the experiments can be performed for a range of assumptions to generate organization-neutral implications. Much of the art of system dynamics modeling is discovering and representing the feedback processes [54]. The next step involves moving the causal diagram to a formal representation of the model. It entails describing detailed relationships between causal links. Nonlinearities and discontinuities in the relationships can easily be incorporated to ensure that the model adequately reflects reality. Application to a variety of organizational and environmental conditions rounds out model development. When applying the model, the emphasis is on analyzing

Table 1. Systems Dynamics Notation

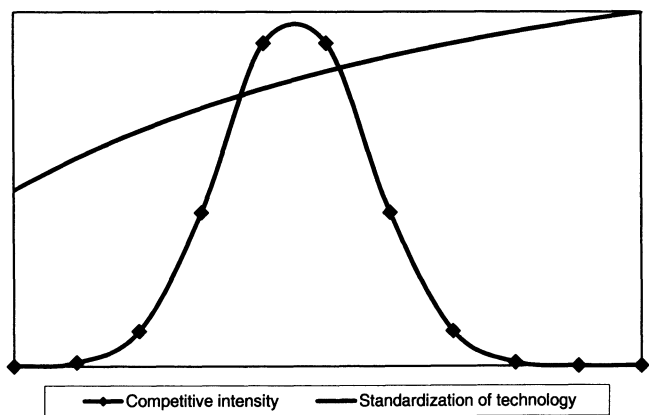
Concept	Explanation	Notation
Stock	Represents an accumulation. Includes physical and abstract variables.	Stock
Flow	Connects a stock and causes changes in the level of the stock.	
Connector	Conveys information.	
Converter	Holds values of input, output, and intermediate variables. Does not accumulate.	<i>Variable_Name</i>

behavior patterns, trends, and how quickly a variable achieves stability, rather than on absolute values generated through the simulation. Recalibration of the model equations and parameters will permit us to focus on the actual numbers, although model calibration for a specific firm is not the part of current research.

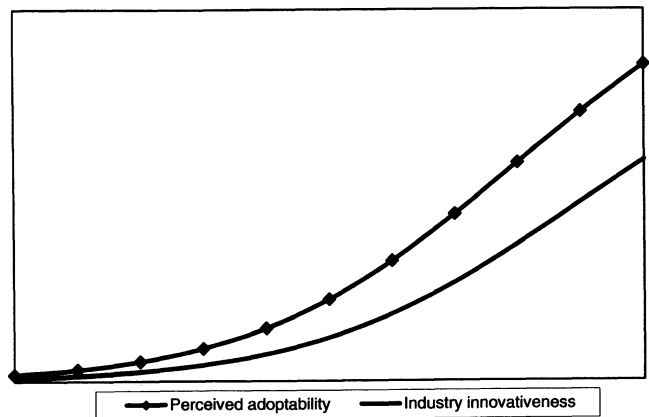
The Adoption Model Adapted for SOA

MULTIPLE APPROACHES HAVE BEEN EMPLOYED to study the adoption phenomenon, including studies at the industry level [6] and from an individual firm’s perspective [31, 49]. In much of the prior research, the innovation is frequently treated as a black box. This research posits that SOA has some unique properties that change the way a firm operates and interacts with its environment, and these need to be considered in the adoption decision. Specifically, SOA provides an organization the opportunity to quickly change applications in response to changing business needs and affords greater IT–business alignment, potentially increasing business value. These factors are incorporated into the model, which is presented in Figure 1. A detailed description of the model follows. It should be acknowledged that while the model was crafted to understand the SOA adoption process, much of its theoretical underpinnings are based on general technology adoption research, which is SOA-neutral. Some constructs are specific to SOA adoption, such as the availability of services. However, a majority of the constructs are applicable to any new IT. As a consequence, with suitable recalibration, the model can be applied to other innovative ITs. Some factors that are unique to SOA may need to be dropped, and others specific to other technologies may need to be introduced.

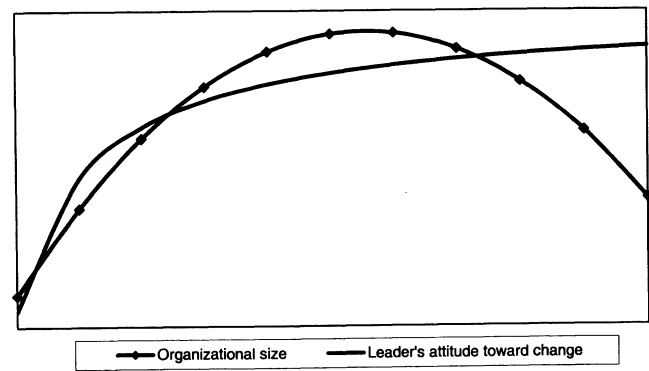
The upper left portion of the model covers industry diffusion and the bottom right segment addresses individual firm adoption processes. At the industry level, diffusion is determined by the difference between the adoption and discontinuance rate, moving firms between two interrelated stocks, *Potential Adopters* and *Adopters*. Movement from one to another is moderated by the *Adoption_Rate* and the *Discontinuance_Rate*.



(a) Determinants of industry innovativeness



(b) Determinants of adoption rate



(c) Determinants of organizational innovativeness

Figure 2. Model Calibration

applied to a different firm, industry, context, or technology, these relationships will have to be recalibrated.

In the context of this study, *Industry_Innovativeness* represents the degree to which an industry is willing to embrace new ITs. It should be stressed that this is not a measure of how many innovations have been developed by the industry over time. Because the firms are not likely to be creating SOA products, the model uses industry innovativeness to characterize their propensity to adopt new IT. Industries that are dependent on IT for competitive advantage are more likely to embrace new IT solutions. Industry innovativeness is a function of competitive intensity and the standardization of technology in that industry. As the level of competitive intensity rises, so does the industry innovativeness. However, beyond a certain point, innovativeness is stifled due to resource depletion [48]. The relationship can be configured for any industry, changing the height and shape of the curve. Industry innovativeness is also affected by the level of technology standardization with increasing standardization leading to greater propensity for the industry to adopt an innovation [48], though at a decreasing rate. These relationships are depicted in Figure 2a, where the y-axis represents industry readiness as a function of competitive intensity and standardization of technology. The representation permits viewing the separate effects of the two drivers of industry innovativeness at the same time. A three-dimensional representation was initially employed, but the separate effects were harder to interpret. The same logic has been used for Figures 2b and 2c.

Perceived adoptability of SOA is determined using a subset of Rogers's characteristics of innovation that are most relevant to IT innovations—namely, relative advantage, complexity, and compatibility [27, 43, 49, 50]. The strong linkage between relative advantage and adoption rate is well documented [49]. *Relative_Advantage_of_SOA* represents the degree to which SOA is perceived as being better than the ITs it replaces. Although compatibility and complexity are comparatively less important in predicting adoption rate, they are positively related to adoption rate as well. *SOA_Compatibility* represents the degree to which SOA is perceived as consistent with the existing IT architecture and infrastructure, coupled with the ability to meet the needs for the IS that support the business. Similarly, *SOA_Complexity* is the degree to which SOA is perceived as relatively difficult to implement and use. As the literature suggests, increased perceived adoptability leads to an increasingly higher adoption rate. In a similar manner, industry innovativeness will also have a positive effect on SOA adoption, though the effect is not as pronounced, as depicted in Figure 2b.

The adoption of any innovation within an industry is the sum total of a number of individual adoption decisions. Each of these decisions is shaped by a number of factors, including the technology appropriateness, organizational innovativeness, and the extent to which the innovation has already been adopted in the industry. In addition, in the case of SOA adoption, the magnitude of IT–business discrepancy also affects the decision to adopt. These factors represent the theoretical constructs of task–technology fit [14]; attitude toward change [49]; peer pressure in the form of coercive, mimetic, and normative pressures [35, 57]; and IT–business alignment [10]. These factors do not equally contribute over the adoption process. In the early stages of industry-wide

diffusion, organizational innovativeness [49] and task–technology fit [36] are more relevant to a firm’s adoption decision. Later on, normative and political pressure is likely to be more appropriate [14].

To accommodate these factors, the model employs multiple determinants for the adoption decision. The percentage of adopters serves as a moderator, allowing organizational innovativeness to play a major role during early stages of diffusion, SOA appropriateness (representing task–technology fit) during early maturity stages, and peer pressures during late maturity stages. The degree of organizational innovativeness is shaped by the organization’s size and the attitude of its leaders toward change—with very small and very large organizations considered less innovative [31], and leader’s attitude toward change providing a positive influence though attaining saturation after a point, as illustrated in Figure 2c.

SOA_Appropriateness is shaped by the extent to which SOA can deliver the required performance and security needed by the organization’s applications [33]. Some SOA-specific decision factors are introduced—namely, the availability of services and SOA cost-effectiveness. The development of a services marketplace and availability of appropriate services for the domain is essential for successful and cost-effective SOA adoption. This was borne out in the case of component-based software development, wherein immature markets tended to discourage adoption of the technology. Likewise, the relative cost-effectiveness of SOA over traditional monolithic technologies to develop, deliver, operate, and maintain applications is a prominent driver of SOA adoption [12]. While IS agility enabled by SOA can arguably affect SOA appropriateness, we opted to model these as separate determinants of the decision to adopt SOA.

The other segments of the model deal with the impact of adopting SOA on the organization. As changing business processes widen the IT–business discrepancy, the firm’s intention to adopt agile IT such as SOA increases. The flexible and dynamically reconfigurable end-to-end business processes realization of SOA enhances IS agility [10]. This in turn will give rise to implementation activity to support the changed business process. At this stage, an appropriate SOA governance mechanism needs to be developed and implemented. In the case of SOA adoption, the learning curve and the introduction of the governance mechanism will introduce a delay in implementation as opposed to using current technology. However, the flexibility afforded by SOA, coupled with the ability to reuse components, will likely lead to faster IS delivery over traditional IS technology. Detailed specifications of all model constructs are included in the Appendix, which shows the mathematical relationships between constructs, constants employed, and initial values of accumulating constructs. Relationships that are specific to a firm, industry, context, or technology appear in boldface.

Simulation Results

THE SIMULATION WAS CONDUCTED USING Vensim® PLE (Personal Learning Edition), a fully functional system dynamics software package from Ventana Systems Inc. It was run over a period of 60 months, representing a medium-term planning horizon. It was tempting to simulate for longer terms, but the uncertainty of environmental conditions over an extended period precludes making meaningful assessments and predictions.

The experiments are conducted with two objectives—to validate that the model is performing realistically and to understand the impact of the adoption decision under different environmental conditions.

Model Validation

Design science research strongly stresses the need to evaluate and validate the design artifact [32]. Several mechanisms are recommended, including observation, analytical methods, simulation and experiments, testing, and descriptive persuasion. The choice of technique is based on the type of the design artifact and the nature of evaluation to be performed. Some candidate applications include observation for case and field studies, analytical methods for artifacts involving deterministic models, simulation and experiments for artifacts incorporating nondeterministic or stochastic models, testing for software artifacts, and descriptive persuasions for other artifacts. For the artifact at hand—a system dynamics model of SOA adoption—simulation represents the preferred validation technique. Validating a simulation model typically involves demonstrating that the model results are consistent with the real world [25].

Validation of system dynamics models is performed using structural testing and behavioral pattern testing [24]. Structural tests seek to ensure that the equations used in formulating the model are consistent with empirical or theoretical knowledge about the real-world system. These tests are static in nature and involve no execution of the model. Behavioral pattern tests, on the other hand, seek to determine if the results from execution of the simulation model are consistent with patterns observed in the real-world system. These are dynamic tests and emphasize predictions of patterns (frequencies, trends, etc.) as opposed to individual point predictions. The construction of a system dynamics model is an iterative process, with successive application of structural and behavioral pattern testing. Failure, or unexplained performance, on either validation strategy will necessitate revision of the model equations and structure.

The SOA adoption model was validated using both approaches. Structural testing examined the model constructs and equations. Constructs in the model were derived from existing theory and research in adoption of ITs. The equations were derived using findings from adoption and diffusion research. In a further effort to demonstrate soundness of the model, the equations were subjected to extreme value and boundary value analysis under a variety of conditions to detect any brittleness in the model. Any unusual patterns were flagged for further investigation and recalibration if necessary. After considerable structural testing, the model was deemed robust enough for behavioral pattern testing.

The adoption model in this research was unique in that it addressed adoption at the industry and the organization levels simultaneously. Behavioral pattern testing would need to validate that the model's results are consistent with real-world findings at both levels concurrently. At the industry level, the adoption rate reflects a balance between adoption and discontinuance and is expected to follow an inverted U-curve. Because the adoption rate is dependent on industry innovativeness and perceived adoptability, increasing competitive intensity causes industry innovativeness to erode. The resulting effect is a classic S-shaped curve in cumulative adopters [29]. The results of the

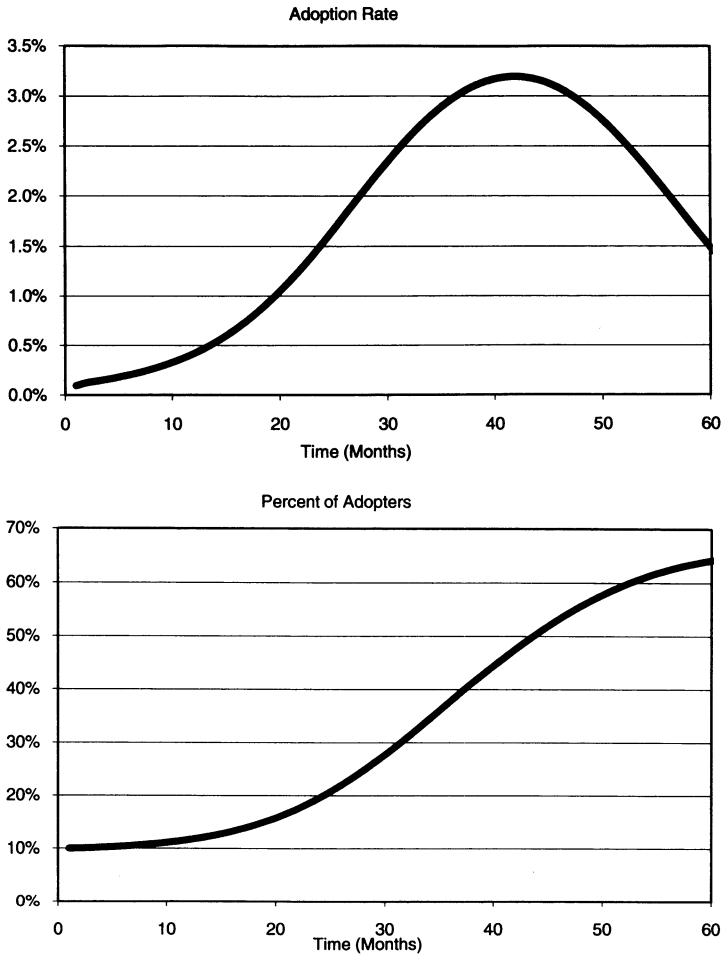


Figure 3. Industry-Level Simulation Results

behavior pattern testing confirm this, and appear in Figure 3 depicting monthly and cumulative adoptions as a percentage of the total industry population. The resulting patterns confirm that the model follows theoretical expectation.

At the organizational level, a change in the underlying business processes should lead to an increase in the amount of IS implementation activity, which in turn will cause changes in the portfolio of IS, and thereby reduce the discrepancy between business processes and the technology that supports them. If SOA is adopted, then the responses should be relatively quick. This is illustrated in Figure 4, with the y-axes representing changes to the number of processes and IS. The simulation entails changes to business processes at specific points in time based on external environmental changes. This immediately triggers a discrepancy between the existing IT portfolio and the business needs. As a result, the organization seeks to modify some of its application systems. This is captured as implementation activity. When SOA is initially

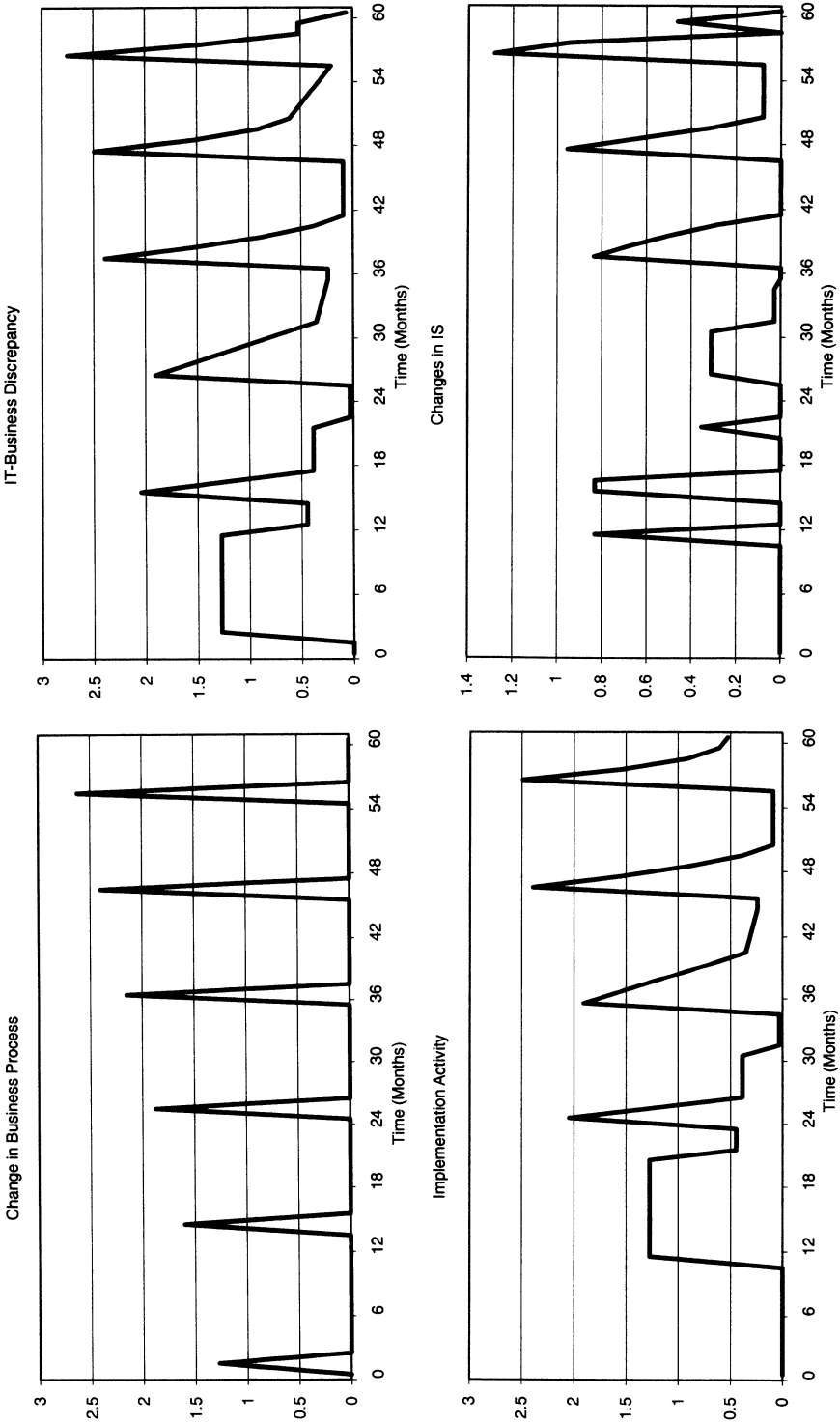


Figure 4. Organizational-Level Simulation Results

adopted, the implementation activity is lagged and not quite as effective. Over time, as IT personnel acquire greater competence, the delay in implementation is reduced and the effectiveness of this activity is greater. The implementation activity leads to changes in IS, which in turn reduces the IT–business discrepancy. Once again, the model performance is consistent with expectations, further reinforcing successful behavioral pattern validation [24].

Scenario Analysis

The results from the previous section serve to demonstrate that the model is functioning competently. It generates results that are consistent with expectations, and the performance is stable with no wild fluctuations noted. However, these results shed light on only one set of conditions—the experimental conditions chosen for this simulation—representing a narrow slice of possible scenarios that an organization may face when contemplating and implementing SOA. For managers seeking to adopt SOA, greater insight is sorely needed, given that the investment is nontrivial and entails significant retraining of existing IS personnel and, possibly, new recruitment. A more comprehensive exploration of the various conditions is needed. This will also demonstrate the robustness and fidelity of the model. A systematic exploration of the problem space will also improve generalizability of the findings, while proving invaluable to managers. Accordingly, the model was evaluated for a number of different conditions and scenarios. The results were consistent with expectations, though some of the scenarios proved less insightful than others. A set of alternative scenarios is examined to illustrate the effect of changing model parameters and the implications for SOA adoption.

Contrasting SOA Adoption and Nonadoption

For a manager looking to adopt SOA, an analysis of alternative scenarios involving SOA and traditional monolithic IT would prove useful. This is accomplished by manipulating the nature of the environment competitiveness, wherein SOA will be adopted more readily in competitive environments. Simulations were run for these scenarios, with the resulting impact on IT–business discrepancy and changes to IS presented in Figure 5.

The results for SOA-based implementation are illustrated with a solid line and that of legacy architectures appear using a dashed line. Several observations are readily apparent. Modifications to IS occur more slowly and in smaller magnitude for traditional IT as compared to SOA. In addition, the area under the curve is less for SOA, indicating that the total effort to modify the relevant applications is less than in the case of legacy architecture. More importantly, the graph illustrates the potential for problems in a more volatile business environment. If the changes occur more quickly than is currently the case, the maintenance effort will snowball, resulting in delays to the modification of applications. This does not apply for SOA, due to the quicker response to request for change. The impact on IT–business discrepancy is more pronounced. The duration

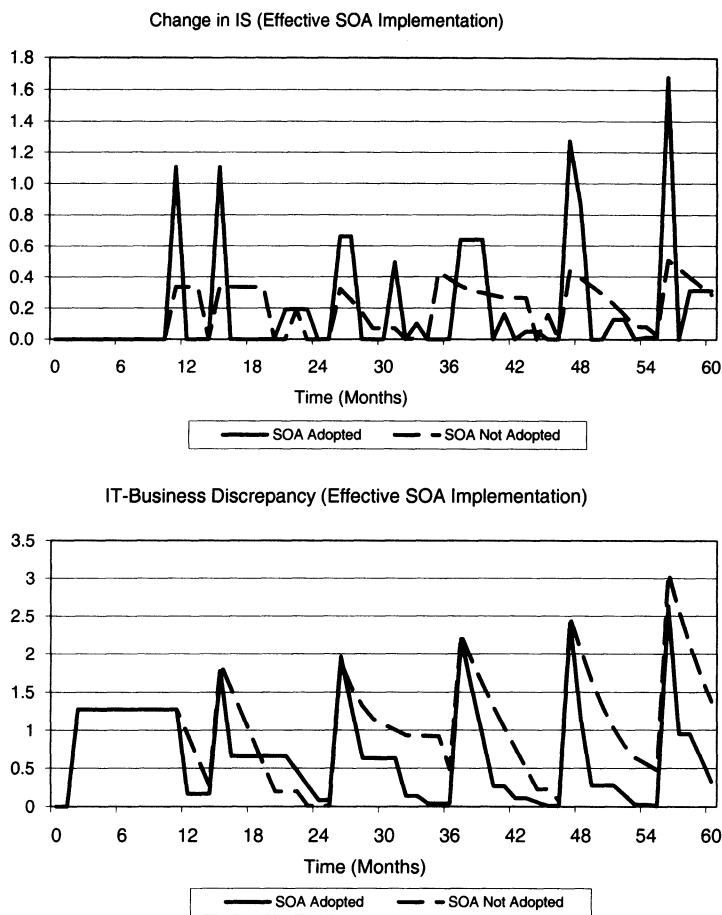


Figure 5. Benefits of Adopting SOA

and magnitude of difference between the business processes and the IS that supports them is far less for SOA than for legacy architectures. Moreover, the transition from a spiked response to a sawtooth response leaves the organization vulnerable to snowballing problems in more volatile conditions. This can be observed toward the end of the simulation, where the discrepancy is zero for the SOA case, while it is perceptibly greater with each change cycle in the legacy case. The response curves will vary by industry and organization, but the overall trend is expected to be similar.

The Impact of Organizational Readiness

Although the relationship between organizational size, leader’s attitude toward change, and organizational innovativeness is well documented [16, 31, 49], its eventual impact on an organization’s intention to adopt SOA is not so clear. This set of simulation exercises examines the impact of organizational readiness on SOA adoption. Given that organizational size and leader’s attitude toward change have nonlinear effects,

Table 2. Organizational Readiness Simulation Scenarios

Organizational readiness	Leader's attitude toward change	
	Receptive	Resistant
Organizational size		
Large		Medium (case 2)
Medium	High (case 1)	
Small		Low (case 3)

we sought to differentiate among alternative levels of organizational readiness by considering multiple conditions generated through three levels of organizational size and two levels of leader's attitude toward change, as illustrated in Table 2.

Although there are six possible scenarios to consider, we sought to examine the effect of contrasting levels of organizational readiness on SOA adoption. Consequently, we selected a case of high readiness, characterized by a medium-sized firm with a receptive attitude (case 1); a case of moderate readiness embodied as a large firm with a resistant attitude (case 2); and a case of low readiness, represented by a small firm with a resistant attitude (case 3). The results appear in Figure 6.

The intention to adopt SOA is consistently highest for the organization with the greatest readiness (depicted as a dotted line in Figure 6), while it is noticeably lower for the organizations considered moderately ready (solid line) and those considered not ready (dashed line). Discontinuities in the curves at months 24 and 51 arise from industry-level effects as the percentage of adopters increase, with organizational innovativeness playing a major role during early stages of diffusion, technology appropriateness during early maturity stages, and peer pressures during late maturity stages, as discussed in the fourth section. As the first transition point is reached, the intention to adopt drops for cases 1 and 2 due to the diminished effect of technology appropriateness. This reduction is much more pronounced for case 1 than case 2, where the overall level of intention to adopt SOA is lower, suggesting that organizations with higher innovativeness may be more inclined to SOA discontinuance when technology appropriateness decreases. This is understandable because organizations with higher innovativeness also tend to be more open to change. In the case of organizations not considered innovative (case 3), this transition is marked by a step up in the intention to adopt, as the perception of greater technology appropriateness plays a more significant role in the decision. After the initial discontinuity, the intention to adopt SOA starts to rise gradually due to an overwhelming increase in the percentage of adopters. The primary weight factor, technology appropriateness, cannot completely offset this rising trend. The three cases are characterized by similar trends. Toward the end of the simulation, another discontinuity is observed. This is characterized by a sharp decrease, reflecting the diminishing pattern of technology appropriateness. Although the primary weight factor in these stages is percentage of adopters who are already in the highest value range, the overall level of intention to adopt SOA drops due to the reduction of technology appropriateness and marginal gain in percentage of adopters.

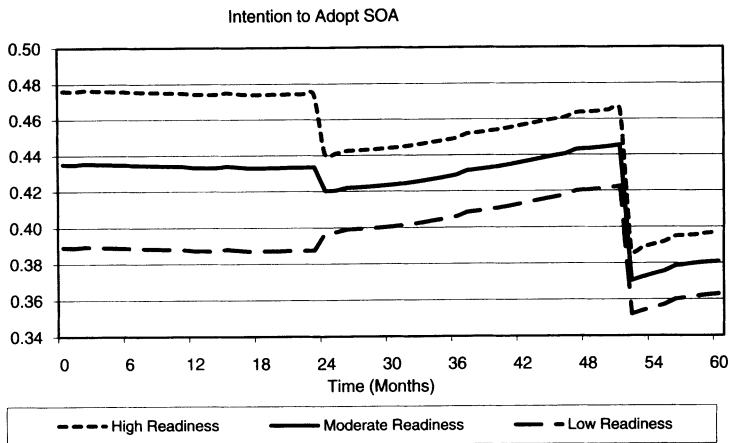


Figure 6. The Effect of Organizational Readiness

The Impact of Industry Readiness

Industry-wide diffusion of SOA is shaped by many conditions, rendering normative pressure on individual organizations to adopt an innovation [57]. This set of simulations seeks to examine the impact of industry readiness on the decision to adopt SOA. Once again, contrasting levels of industry readiness were simulated by selecting different levels of SOA standardization and the availability of alternative IT infrastructure frameworks, as illustrated in Table 3.

Although the simulation conditions generate four possible cells, we sought to examine the high and low cases of industry readiness only. Higher levels of availability of alternative IT architecture frameworks can result in sluggish diffusion of SOA technology, especially when coupled with low standardization of the technology. On the other hand, the speed of SOA diffusion can be enhanced by reasonable standardization of the technology [48], and this is enhanced when there are fewer competing alternative technologies. The results are depicted in Figures 7 and 8.

It is apparent that industry readiness plays a major role in the decision to adopt SOA. Low industry readiness, as characterized by low standardization of technology and availability of several alternative technologies, results in a relatively flat curve with only one discontinuity. As experienced earlier in the case of organizational readiness, this transition represents the shift from organizational innovativeness to technology appropriateness. However, the low intention to adopt ensures that the number of adopters remains low, and as a result the second transition is never experienced during the course of the simulation. High industry readiness, on the other hand, presents a rich expanse of adoption behavior. As the number of adopters increases, the threshold for the first transition is reached earlier (in month 14, as opposed to month 38 in the low industry readiness case), and the switch from organizational innovativeness to technology appropriateness results in a drop in the intention to adopt SOA, as characterized in the earlier case. Given the increased propensity to adopt innovations, the curve starts to

Table 3. Industry Readiness Simulation Scenarios

Industry readiness	Availability of alternative technology	
	High	Low
Standardization of technology		
High		High industry readiness
Low	Low industry readiness	

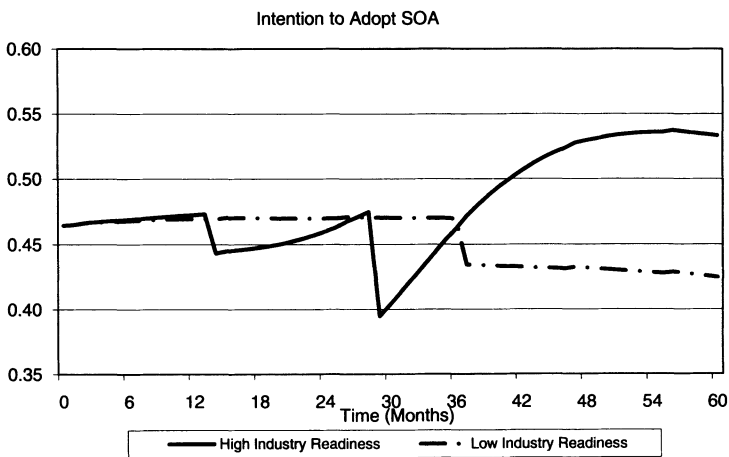


Figure 7. The Effect of Industry Readiness

pick back up rapidly. The large number of adopters leads to a second transition, wherein the major effect switches from technology appropriateness to normative pressures—generating the second discontinuity. After this point, the percentage of adopters is the major driver, and given the speed of SOA diffusion within the receptive industry, the curve picks up rapidly. Toward the end of the simulation, some discontinuance of adoption is experienced, and as a result the curve dips to some extent.

The effect of industry readiness on organization IT infrastructure is also worth examining. In this case, we presented the results relating to IT–business discrepancy and the changes to IS. With an adoption threshold of 0.48, both high and low industry readiness cases tend not to adopt SOA until month 38, when only the high industry readiness curve exceeds the SOA adoption threshold. As a result, the curves in both cases are identical, reflecting the reliance on traditional monolithic IT infrastructure. From month 38 onward, SOA is implemented and deployed only in the high industry readiness case, leading to distinct curves for the rest of the simulation. Modifications to IS occur more slowly and in smaller magnitude for the low industry readiness case. In addition, the modification activity is nearly continuous. As a contrast, the high industry readiness situation embodies more intense and more effective changes to IS, with little snowballing of implementation activity. This is further reflected in the IT–business

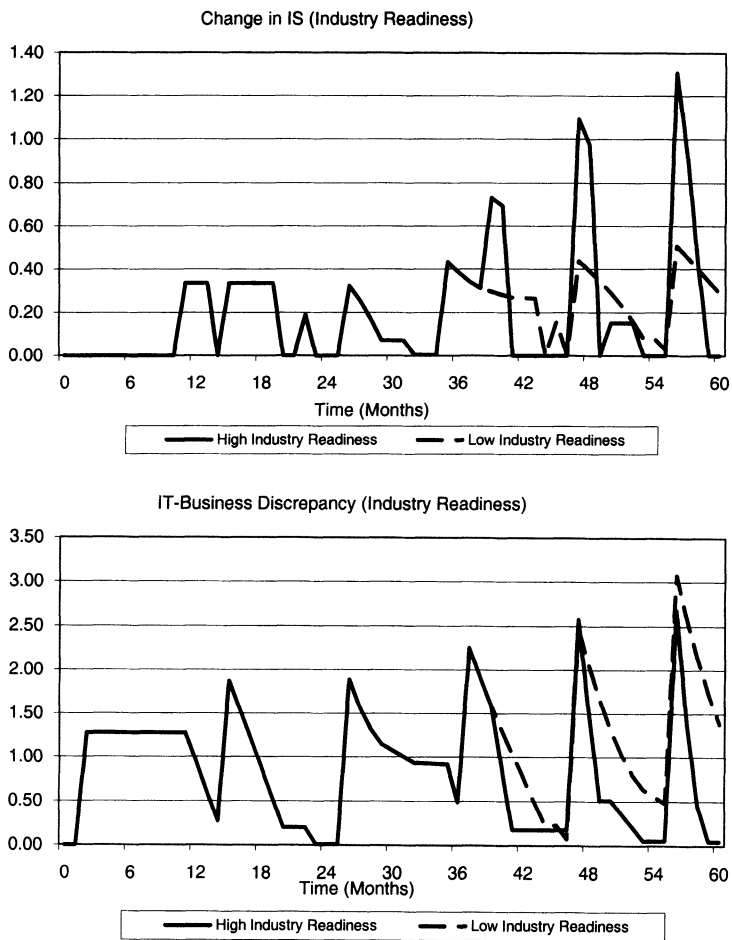


Figure 8. Industry Readiness Effect on Organization IT

discrepancy graphs, wherein the duration and magnitude of difference between the business process and the IS that supports it is far less for the high industry readiness case than for the low industry readiness case. As seen before, the benefits of adopting SOA cannot be overstated, even under different environmental conditions.

The Impact of SOA Implementation Effectiveness

From the simulations performed so far, it would appear that adopting SOA would be a preferable decision under most circumstances—delivering the promised benefits over traditional monolithic IT alternatives. However, like any new IT, SOA is also subject to missteps and has its fair share of failed deployments [60]. Most of these arise from inappropriate migration approaches [40] and the lack of a suitable SOA governance model [11, 60], implying that many organizations are not able to effectively identify, isolate, and integrate services with legacy systems for maximum reusability. As a

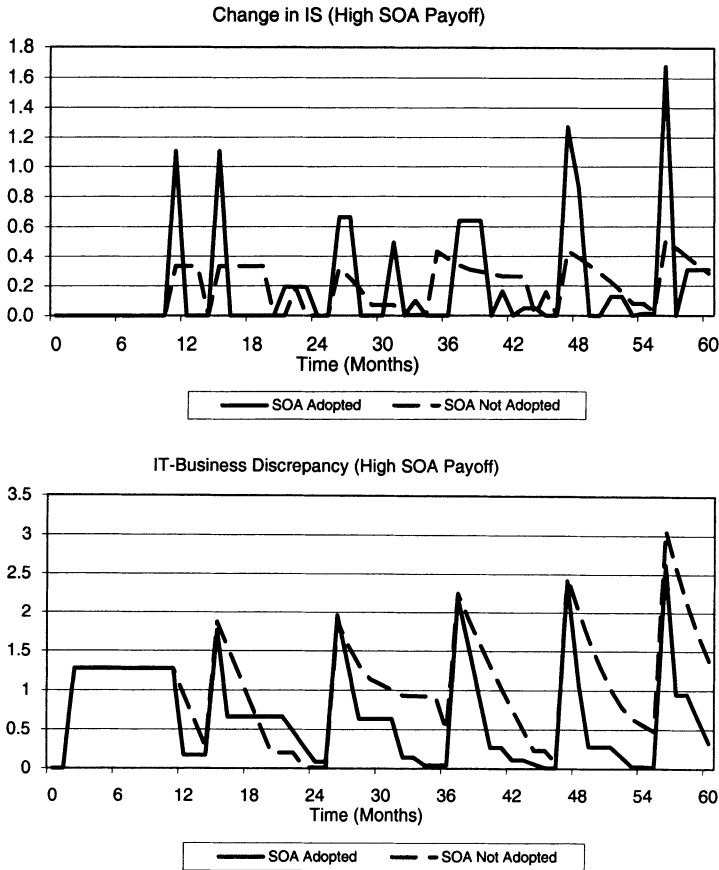


Figure 9. The Effect of High SOA Effectiveness

consequence, it is not surprising that companies are struggling to see the realization of SOA benefit [30]. In an attempt to understand how the relevant effectiveness of resulting SOA-based infrastructure plays role in the business value, a further set of simulations was performed. In this case, we manipulated a coefficient governing the effectiveness of system modification. These results are depicted in Figures 9 and 10.

The graphs contrast the effect of adopting SOA versus nonadoption in cases of high SOA payoff. The changes in IS occur more quickly and more effectively when the SOA deployment is effective. Changes using the traditional IT infrastructure are more continuous, which introduces the need for some caution, because this is susceptible to snowballing if business changes are more rapid. The IT–business discrepancy in Figure 9 is more telling, with longer periods of greater discrepancy in the nonadoption case. As before, this confirms expectations regarding the promise of SOA when deployed effectively.

However, if SOA is not deployed effectively, the results are not quite as encouraging. In this case, the organization flounders as it seeks to implement the new technology, and the effort involved in making changes to IS may even exceed the case of traditional

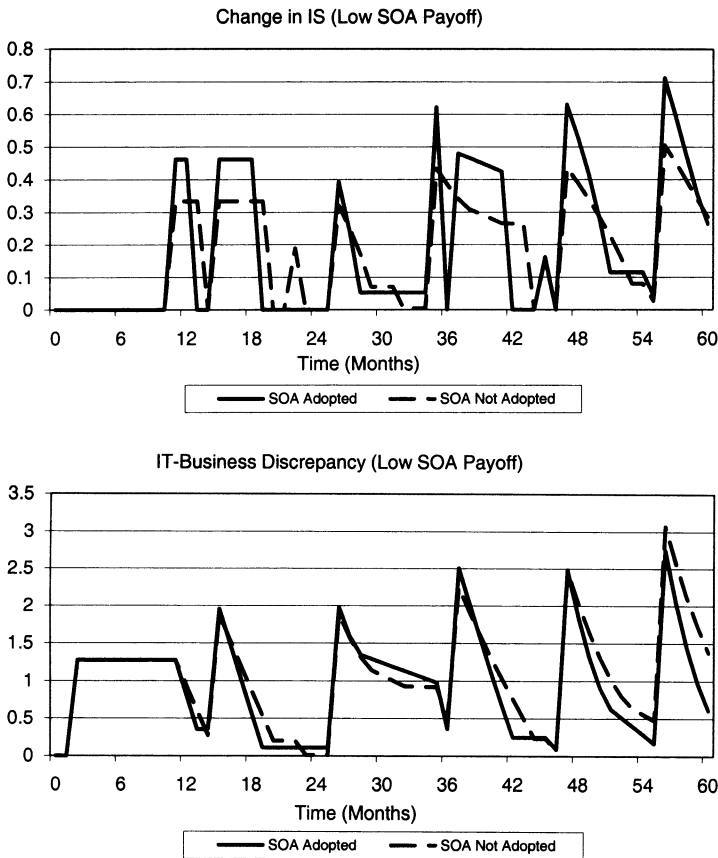


Figure 10. The Effect of Low SOA Effectiveness

IT, where familiarity and prior knowledge will clearly offset the purported benefits from SOA, coupled with the learning curve for implementing SOA. The IT–business discrepancy is very much the same for the two cases, suggesting that ineffective SOA implementation does not provide the organization with expected benefits. Given the nontrivial cost of introducing SOA into an organization, it is apparent that this can lead to discontinuance of the innovation, as is anecdotally reported [30].

Most of the results were consistent with expectations. From a behavior reproduction test perspective, this confirms that the model performed satisfactorily for the scenarios explored. However, there are a few unexpected outcomes. These need to be examined in greater detail. At month 15, the difference between the business process and the IS that supports it begins to fall in both SOA and non-SOA cases. While IT–business discrepancy starts to close more quickly with SOA technology, the closing behavior abruptly stops at month 16 and remains static until month 21, allowing discrepancy in the non-SOA case to stay less than in the SOA case until month 25. Upon closer scrutiny, it transpires that modification activities to relevant IS are driven down to zero, rendering the discrepancy curve flat for the period. This is indicative of

a case where the organization realizes that IS modification through a newly planned SOA-based system overshoots the goal of minimizing IT–business discrepancy and therefore pulls the plug on planned implementation. Moreover, SOA-based systems may entail a significant delay before launching implementation activity, as it takes longer to design SOA and supply requisite resources to implement SOA. In this case, when the planned implementation is ready to be carried out, the assumption of wider IT–business discrepancy no longer holds. The results indicate that a delay of SOA implementation has a potential to seriously erode the attractiveness of SOA, let alone lower SOA implementation effectiveness.

Managerial and Research Implications

THE MODEL AND THE ACCOMPANYING SIMULATIONS ILLUSTRATE that the adoption decision and implementation strategies have definite implications for effective IS modification leading to IS agility. A number of clear inferences can be deduced through analysis of various scenarios using the model. In a stable environment, traditional IT architecture performs relatively well, rapidly closing the gap between business processes and the applications supporting them. Under these conditions, the advantage of an SOA-based infrastructure is not nearly as prominent as claimed, especially when assessed in terms of IT–business discrepancy, though the overall effort to modify the relevant applications is still significantly less in the case of SOA. This suggests that cost savings rather than performance enhancement would be the primary driver for SOA adoption in a stable environment. In a more volatile environment, one that is characterized by a greater competitive intensity and changes in business process, similar trends are observable in the modification activities to IS applications, indicating that the cost-saving advantage of SOA still applies. Moreover, the gap between business processes and IS support for them closes much faster with an SOA-based infrastructure. Clearly, SOA has some significant advantage in this case, more so if the changes to business processes occur more rapidly and with greater magnitude.

The results also suggest that organizations that are more innovative are more likely to implement SOA and are more likely to be successful in this endeavor. However, when faced with more demanding security and performance requirements, the underlying appropriateness of SOA starts to erode, and innovative organizations are prone to discontinuing the use of the technology. Less-innovative organizations, though slower to adopt SOA, are more likely to continue using SOA for a longer period, even in the face of eroding technology appropriateness. Although seemingly counterintuitive, an examination of the simulation data yields clear explanations for this. Innovative organizations are more receptive to change and are likely to recognize that decreasing technology appropriateness calls for discontinuance of the technology rather than further investment. Although technology appropriateness does serve as a facilitating factor for SOA adoption, it does not apply universally.

In a more mature industry that is open to the use of SOA technology, the decision to adopt SOA is shaped by the perceived superior business value and tends to occur during the later stages of simulation. The results obtained by varying the industry environ-

Table 4. Summary of Simulation Scenarios

Scenario	Changes to IS (total number of changes)	IT–business discrepancy (cumulative count of misaligned processes over simulation)
SOA adopted	11.91	41.71
SOA not adopted	10.83	65.10
High industry readiness	11.87	51.52
Low industry readiness	10.83	65.10
High organization readiness	11.86	46.01
Medium organization readiness	11.39	50.73
Low organization readiness	10.49	51.77
SOA adopted: effective deployment	11.91	41.71
SOA not adopted: effective deployment	10.83	65.10
SOA adopted: ineffective deployment	11.58	58.63
SOA not adopted: ineffective deployment	10.83	65.10

ment conditions served to confirm that the model performed correctly for a variety of conditions, though it did not yield any additional insights from those gleaned through prior simulations. The ability to control and systematically vary modeling conditions representing different scenarios permits managers to explore different industry conditions and examine the impact on a variety of variables of interest.

These findings are summarized in Table 4, wherein the changes made to the IS portfolio and the cumulative IT–business discrepancy are presented over the entire simulation for all scenarios described. Changes to IS represents the number of IS changes over the simulation period—note that this is not an integer, because changes can be made to a part of an IS, and the change could span several months, leading to fractional scores. This measure is indicative of the maintenance effort expended under different scenarios. There is no *a priori* direction of preference in this case, other than it should be driven by the number of changes to underlying business processes. The IT–business discrepancy measure represents the cumulative count of misaligned processes over the course of the simulation. Larger values on this core are indicative of slower responses to changes in the business processes and lower levels of agility. The values for this measure are considerably higher than the changes to IS measure because it often takes several months to effect a change, and the process is considered misaligned during the entire period. As before, fractional values are possible, indicating partial misalignment between process and supporting IS. Some caution needs to be exhibited when interpreting these numbers—they are less indicative of predictive accuracy and are more suggestive of general trends.

A couple of observations are readily apparent. The number of changes to IS across all scenarios is relatively similar. This reflects the organization's response to changes in business processes, and in all cases, whether SOA was adopted or not, the organization

made a good faith effort to respond. However, the more dynamic environment (higher industry readiness, higher organization readiness, and more agile technology) leads to more changes to IS. The cumulative IT–business discrepancy measure also provides some insight into the different scenarios. The greater separation between these scores is illustrative of the differences between SOA adoption and nonadoption under a variety of conditions. As can be seen in the first segment of the table, SOA adoption gives an organization an opportunity to modify IS more quickly, leading to closer alignment between the business process and its supporting IS. For industries that are able to assimilate SOA more readily, the difference is also nontrivial, though not as pronounced. At the organizational level, a similar trend was noted, moving across high to medium to low levels of readiness. However, in this case, the effect was considerably smaller. The last scenario provided some sobering insights. It demonstrated that, like other ITs, SOA is not a silver bullet, and that its mere adoption will not guarantee the claimed benefits. Ineffective implementation leads to long periods of misaligned business processes and supporting IS. When taken as a whole, the clear implication is that SOA does enhance an organization's ability to respond quickly to changes, but the environmental and organizational conditions must be suitably aligned for improved agility.

SOA implementations do not always yield the promised benefits. Anecdotally, a number of explanations have been advanced for these shortfalls. In addition, a host of best practices have been proposed to leverage the SOA investment for maximum return on the investment. However, organization and environmental conditions vary widely across each implementation and can be expected to change even during SOA implementation. A series of simulations under varying industry and organization conditions provides insight into the key role of SOA implementation effectiveness in realizing business value from SOA investment. In situations where the SOA implementation is effective, the effort and cost to modify IS within the organization's portfolio is markedly less than that required for traditional monolithic IT alternatives. Moreover, the quicker response through the effective use of SOA provides the IT department with a suitable measure of slack between successive change efforts. As a result, the gap between business processes and IS also closes rapidly. These results indicate that SOA can deliver on promised benefits when used appropriately. However, it is not a panacea, or a sure-fire option under all circumstances. In conditions where the appropriateness of SOA is questionable, or where it is not effectively used, these benefits fade rapidly, and SOA is comparable to existing IT alternatives in terms of effort needed to change existing IS, as well as the longer periods wherein business needs and IS remain mismatched. In addition, delays in implementing SOA also yield similar results, indicating that the window of opportunity for using SOA is not indefinite. It is important to recognize that the rate of business change plays a role in these findings. For conditions involving infrequent change, or when the change is widely spread, the benefits of SOA will still accrue, though the opportunities to realize them are fewer and farther between. On the other hand, if the industry is characterized by rapid change, the benefits from effective SOA implementation are much more pronounced. The tendency of implementation effort to snowball is clearly a risk that is of greater concern for traditional ITs.

The SOA paradigm introduces a number of managerial decisions, ranging from strategic decisions to invest in the technology, to tactical decisions concerning buy-versus-build, to operational issues addressing selection, integration, and implementation of services—all of which represent fertile areas for research. This paper examines some aspects of these decisions, focusing on the adoption decision in the context of a larger diffusion phenomenon and examining its impact on firm and IS agility. Employing a design science research paradigm, we created a detailed model of SOA adoption and its impact on IS modification in light of changing business requirements. Using system dynamics, we simulated a number of different scenarios and sought to provide insight into the effect of alternative decisions under a variety of conditions. The model was perfected through repeated application under a variety of conditions. The results generated from the model were carefully scrutinized and compared to known patterns of adoption and diffusion, confirming that the model worked correctly for the scenarios explored. Nonetheless, as with any modeling approach, there is always room for improvement. The model performs well for the scenarios explored, but universal applicability cannot be guaranteed. Organizations frequently experience unanticipated disruptions that radically alter their mode of operation. These will require model reformulation and recalibration.

However, through the exploration of multiple scenarios, this paper opens a window to considerably more research opportunities. Assumptions and input values could be altered to provide additional insight into SOA diffusion and adoption. Input constructs like the change in business processes could be modified to reflect environment volatility by manipulating the magnitude and frequency of changes. Likewise, assumptions regarding the availability of resources and skills for effective development within an SOA framework, the presence of effective governance for managing and controlling services, and the like, can all be relaxed and simulated. Further extensions could incorporate costs of adoption, development, operation, and maintenance of applications within the SOA framework. A different tack could analyze the staffing issues examining alternative options such as contract personnel, third-party hosting services, and the like. Linking SOA to business value represents another research avenue for exploration.

Conclusions

THIS PAPER DESCRIBES ONGOING RESEARCH to study the dynamics of SOA adoption. It employs a modeling approach that is based on system dynamics, permitting managers to explore the implications of alternative decisions when seeking to adopt and implement SOA. The model permits evaluation of the multifaceted, dynamically changing, and interactive factors that affect SOA adoption. As currently formulated, many variables in the model are dimensionless, facilitating comparison among cases and easier interpretation of results. When applied to a specific organization, this will entail reformulation to include measures with dimensions that are meaningful to the organization, as well as recalibration to conform to the specific relationships observed in that context. In its current configuration, the model can be calibrated to reflect different industry

and organizational contexts. These factors can be manipulated to permit a systematic what-if analysis, thereby serving as an effective decision support system.

The use of system dynamics to explore the adoption of SOA represents a novel and unique contribution. The incorporation of the industry-level and organization-level adoption patterns in the model represents another unique contribution. Most models focus either on industry-wide diffusion patterns or on an organization's decision to adopt new technology. By carefully integrating the two models from different levels, we were able to examine the consequences of an organization's decision to adopt SOA in the context of the large picture of industry-wide diffusion. This opens up the possibility of alternative streams of research to study situations where organizations elect to go with or against the industry trend for SOA adoption. Manipulation of these scenarios will entail some recalibration of the model, though.

As with any model, there is always room for refinement. The model assumes that sufficient skills and resources are available for implementation activity. In a different organization, these skills may be in short supply and may need to be contracted out, or may entail additional hiring. Both of these options entail additional cost, which will likely alter the business case for SOA. For organizations that maintain sustained attention to the bottom line, this may serve as an additional reason for discontinuance. Other limitations include the omission of the type of innovation decision, the nature of the social system, and the extent of change agents' promotion efforts—facets of the adoption decision that have been discussed in the literature [49].

Despite these limitations, the model provides managers with an opportunity to engage in a what-if analysis of many competing scenarios. Although a systematic and complete exploration of all possible combinations of conditions is beyond the scope of this paper, the fundamental tools are available for additional examination. These insights should prove invaluable to managers when assessing the decision to adopt SOA, not just in terms of whether to adopt, but also whether early adoption or later adoption would be a preferable strategy given prevailing organizational and environmental conditions.

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REFERENCES

1. Abdel-Hamid, T.K. The dynamics of software project staffing: A system dynamics based simulation approach. *IEEE Transactions on Software Engineering*, 15, 2 (February 1989), 109–119.
2. Abdel-Hamid, T.K.; Sengupta, K.; and Swett, C. The impact of goals on software project management: An experimental investigation. *MIS Quarterly*, 23, 4 (December 1999), 531–555.
3. Arsanjani, A.; Borges, B.; and Holley, K. Service-oriented architecture: Components and modeling can make the difference. *Web Services Journal*, 9, 1 (2004), 34–38.
4. Attewell, P. Technology diffusion and organizational learning: The case of business computing. *Organization Science*, 3, 1 (February 1992), 1–19.
5. Bardhan, I.; Demirkan, H.; Kannan, P.K.; Kauffman, R.J.; and Sougstad, R. An interdisciplinary perspective on IT services management and service science. *Journal of Management Information Systems*, 26, 4 (Spring 2010), 13–64, this issue.

6. Bass, F.M. A new product growth model for consumer durables. *Management Science*, 13, 5 (May 1969), 215–227.
7. Bell, M. *Service-Oriented Modeling: Service Analysis, Design, and Architecture*. New York: Wiley, 2008.
8. Benazeer, M.S.; Verelst, J.; Grembergen, V.W.; and Mannaert, H. Aligning technology with business: An analysis of the impact of SOA on outsourcing. *Journal of Theoretical and Applied Information Technology*, 4, 3 (2008), 244–252.
9. Benbasat, I., and Zmud, R.W. Empirical research in information systems: The practice of relevance. *MIS Quarterly*, 23, 1 (March 1999), 3–16.
10. Bieberstein, N.; Bose, S.; Walker, L.; and Lynch, A. Impact of service oriented architecture on enterprise systems, organizational structures, and individuals. *IBM Systems Journal*, 44, 4 (2005), 691–707.
11. Bieberstein, N.; Bose, S.; Flammante, M.; Jones, K.; and Shah, R. *Service Oriented Architecture (SOA) Compass*. Boston: IBM Press, 2006.
12. Castro-Leon, E.; He, J.; and Chang, M. Scaling down SOA to small businesses. In K.-M. Chao, K.-J. Lin, and W.-H. Wang (eds.), *IEEE International Conference on Service-Oriented Computing and Applications (SOCA 2007)*. Los Alamitos, CA: IEEE Computing Society Press, 2007, pp. 99–106.
13. Choi, J.; Nazareth, D.L.; and Jain, H.K. Modeling the adoption decision process of service-oriented architecture: A systems dynamics approach. In K. Chari and A. Kumar (eds.), *Proceedings of 17th Annual Workshop on Information Technologies and Systems*. Montreal, December 2007, pp. 189–194.
14. Cooper, R.B., and Zmud, R.W. Information technology implementation research: A technological diffusion approach. *Management Science*, 36, 2 (February 1990), 123–139.
15. Coyle, R. The practice of systems dynamics: Milestones, lessons and ideas from 30 years of experience. *Systems Dynamics Review*, 14, 4 (Winter 1998), 343–365.
16. Damanpour, F. Organizational innovation: A meta-analysis of effects of determinants and moderators. *Academy of Management Journal*, 34, 3 (September 1991), 555–590.
17. DiMaggio, P., and Powell, W. The iron cage revisited: Industrial isomorphism and collective rationality in organizational fields. *American Sociological Review*, 48, 2 (April 1983), 147–160.
18. Dutta, A. Business planning for network services: A systems thinking approach. *Information Systems Research*, 12, 3 (September 2001), 260–283.
19. Dutta, A., and Roy, R. Offshore outsourcing: A dynamic causal model of counteracting forces. *Journal of Management Information Systems*, 22, 2 (Fall 2005), 15–35.
20. Dutta, A., and Roy, R. The mechanics of Internet growth: A developing country perspective. *International Journal of Electronic Commerce*, 9, 2 (Winter 2004–5), 143–165.
21. Eisenhardt, K., and Martin, J. Dynamic capabilities: What are they? *Strategic Management Journal*, 21, 10–11 (October–November 2000), 1105–1121.
22. Fichman, R.G. Information technology diffusion: A review of empirical research. In J.I. DeGross, J.D. Becker, and J.J. Elam (eds.), *Proceedings of the 13th International Conference on Information Systems*. Atlanta: Association for Information Systems, 1992, pp. 195–206.
23. Forrester, J.W. *Principles of Systems*. Cambridge, MA: MIT Press, 1980.
24. Forrester, J.W., and Senge, P. Tests for building confidence in system dynamics models. In A.A. Legasto Jr., J.W. Forrester, and T.M. Lyneis (eds.), *System Dynamics, TIMS Studies in the Management Sciences*, vol. 14. New York: Elsevier-North Holland, 1980, pp. 208–228.
25. Gilbert, N., and Troitzsch, K.G. *Simulation for the Social Scientist*. Berkshire, UK: Open University Press, 2005.
26. Glass, R. On design. *IEEE Software*, 16, 2 (March 1999), 103–104.
27. Goldman, K. Perceptions of innovations as predictors of implementation levels: The diffusion of a nationwide health education campaign. *Health Education Quarterly*, 21, 4 (1994), 433–444.
28. Grover, V.; Fiedler, K.; and Teng, J. Empirical evidence on Swanson's tri-core model of information systems innovation. *Information Systems Research*, 8, 3 (September 1997), 273–287.
29. Gurbaxani, V. Diffusion in computing networks: The case of BITNET. *Communications of the ACM*, 33, 12 (December 1990), 65–75.
30. Hau, T.; Ebert, N.; Hochstein, A.; and Brenner, W. Where to start with SOA: Criteria for selecting SOA projects. In R.H. Sprague (ed.), *Proceedings of the 41st Hawaii International*

Conference on System Sciences. Los Alamitos, CA: IEEE Computer Society Press, 2008 (available at www.computer.org/plugins/dl/pdf/proceedings/hicss/2008/3075/00/30750314.pdf?template=1&loginState=1&userData=anonymous-IP%253A%253A71.99.94.194).

31. Haveman, H.A. Organizational size and change: Diversification in the savings and loan industry after deregulation. *Administrative Science Quarterly*, 38, 1 (March 1993), 20–50.

32. Hevner, A.R.; March, S.T.; and Park, J. Design science in information systems research. *MIS Quarterly*, 28, 1 (March 2004), 75–105.

33. Huettenrauch, S. Fundamentals of service-oriented engineering. Working Paper, University of Potsdam, Germany, 2006.

34. Jacobsen, C.; Bronson, R.; and Vekstein, D. A strategy for testing the empirical adequacy of macro-sociological theories. *Journal of Mathematical Sociology*, 15, 2 (1990), 137–148.

35. Katz, M., and Shapiro, C. Technology adoption in the presence of network externalities. *Journal of Political Economy*, 94, 4 (August 1986), 822–841.

36. Laudon, K.C. Environmental and institutional models of system development: A national criminal history system. *Communications of the ACM*, 28, 7 (July 1985), 728–740.

37. Leonard-Barton, D., and Deschamps, I. Managerial influence in the implementation of new technology. *Management Science*, 34, 10 (1988), 1252–1265.

38. Leuthold, F.O. Discontinuance of improved farm innovations by Wisconsin farm operators. Ph.D. dissertation, University of Wisconsin, Madison, 1967.

39. Lewis, G.A.; Morris, E.; Simanta, S.; and Wrage, L. Common misconceptions about service-oriented architecture. In *IEEE Conference on Commercial-Off-the-Shelf (COTS)-Based Software Systems*. Los Alamitos, CA: IEEE Computing Society Press, 2007, pp. 123–130.

40. Li, S.; Huang, S.; Yen, D.C.; and Chang, C. Migrating legacy information systems to Web services architecture. *Journal of Database Management*, 18, 4, (2007), 1–25.

41. Liebrand, W.B.G. Computer modeling and the analysis of complex human behavior: Retrospect and prospects. In W.B.G. Liebrand, A. Nowak, and R. Hegselmann (eds.), *Computer Modeling of Social Processes*. London: Sage, 1998, pp. 1–14.

42. March, S., and Smith, G. Design and natural science research on information technology. *Decision Support Systems*, 15, 4 (1995), 251–266.

43. Moore, G.C., and Benbasat, I. Development of an instrument to measure the perceptions of adopting an information technology innovation. *Information Systems Research*, 2, 3 (September 1991), 192–220.

44. Newcomer, E., and Lomow, G. *Understanding SOA with Web Services*. Boston: Addison Wesley, 2004.

45. Peffers, K.; Tuunanen, T.; Rothenberger, M.A.; and Chatterjee, S. A design science research methodology for information systems research. *Journal of Management Information Systems*, 24, 3 (Winter 2007–8), 45–77.

46. Piccoli, G., and Ives, B. IT-dependent strategic initiatives and sustained competitive advantage: A review and synthesis of the literature. *MIS Quarterly*, 29, 4 (December 2005), 747–776.

47. Roberts, P.W., and Greenwood, R. Integrating transaction cost and institutional theories: Towards a constrained-efficiency framework for understanding organizational design adoption. *Academy of Management Review*, 22, 2 (1997), 346–373.

48. Robertson, T.S., and Gatignon, H. Competitive effects on technology diffusion. *Journal of Marketing*, 50, 3 (July 1986), 1–12.

49. Rogers, E.M. *Diffusion of Innovations*. New York: Free Press, 2003.

50. Rosen, E. *The Anatomy of Buzz: How to Create Word of Mouth Marketing*. New York: Doubleday/Currency, 2000.

51. Sastry, M.A. Problems and paradoxes in a model of punctuated organizational change. *Administrative Science Quarterly*, 42, 2 (June 1997), 237–275.

52. Sharif, A.M. Can systems dynamics be effective in modeling dynamic business systems? *Business Process Management Journal*, 11, 5 (2005), 612–615.

53. Simon, H.A. *The Sciences of the Artificial*. Cambridge, MA: MIT Press, 1996.

54. Sterman, J.D. *Business Dynamics: Systems Thinking and Modeling for a Complex World*. New York: Irwin McGraw-Hill, 2000.

55. Swanson, E.B. Information systems innovation among organizations. *Management Science*, 40, 9 (September 1994), 1069–1092.

56. Swanson, E.B., and Ramiller, N.C. Innovating mindfully with information technology. *MIS Quarterly*, 28, 4 (December 2004), 553–583.
57. Teo, H.H.; Wei, K.K.; and Benbasat, I. Predicting intention to adopt interorganizational linkages: An institutional perspective. *MIS Quarterly*, 27, 1 (March 2003), 19–49.
58. Tornatzky, L., and Fleischer, M. *The Process of Technological Innovation*. Lexington, MA: Lexington Books, 1990.
59. Valente, T.W. Social network thresholds in the diffusion of innovations. *Social Networks*, 18, 1 (January 1996), 69–89.
60. Varadan R.; Channabasavaiah, K.; Simpson, S.; Holley, K.; and Allam, A. Increasing business flexibility and SOA adoption through effective SOA governance. *IBM Systems Journal*, 47, 3 (2008), 473–488.
61. Venkatesh, V.; Morris, M.G.; Davis, G.B.; and Davis, F.D. User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27, 3 (September 2003), 425–478.
62. Wade, M., and Hulland, J. The resource-based view and information systems research: Research, extension, and suggestions for future research. *MIS Quarterly*, 28, 1 (March 2004), 107–142.
63. Westphal, J.D.; Gulati, R.; Shortell, S.M. Customization or conformity? An institutional and network perspective on the content and consequences of TQM adoption. *Administrative Science Quarterly*, 42, 2 (June 1997), 366–394.
64. Zhao, J.L.; Tanniru, M.; and Zhang, L. Services computing as the foundation of enterprise agility: Overview of recent advances and introduction to the special issue. *Information Systems Frontiers*, 9, 1 (March 2007), 1–8.

Appendix: Simulation Details

Equations

Adopters(t) = Adopters($t - 1$) + Adoption – Discontinuance

Adoption = Adoption Rate * Potential Adopters

Adoption Rate: **as per Figure 2b**

Change in Business Process: **pulsed values over simulation time frame as per Figure 4a**

Change in IS: piecewise continuous function with 2 splines for Intention to Adopt SOA and IT–Business Discrepancy

Competitive Intensity = Time * 0.005 + 0.25

Discontinuance = Adopters * (Discontinuance Rate * Implementation Effectiveness)

Discontinuance Rate = 0.05/(20 * Perceived Adoptability + 6)

Industry Innovativeness: **as per Figure 2a**

Industry Performance Requirement = **0.8 * Competitive Intensity + 0.08**

Industry Security Requirement = **1.2 * Competitive Intensity + 0.05**

Intention to Adopt SOA: **piecewise continuous function with 2 splines for Percent of Adopters at 0.2 and 0.6**

IT–Business Discrepancy(t) = IT–Business Discrepancy($t - 1$) + Change in Business Process – Change in IS

Organization Innovativeness: **as per Figure 2c**

Perceived Adoptability = $((0.9 * \text{Relative Advantage of SOA} + 0.1)^2) * ((0.5 * \text{SOA Compatibility} + 0.1)^1) * ((1/(5 * \text{SOA Complexity} + 1))^{0.5})$

Percent of Adopters = Adopters/Total Population

$$\text{SOA Appropriateness} = (1 - 0.4 * \text{EXP}(-1 * 8 * (1 - \text{System Security Requirement}))) * (1 - 0.6 * \text{EXP}(-1 * 6 * (1 - \text{System Performance Requirement})))$$
$$\text{SOA Compatibility} = (30 * \text{Time}) / (10 + 80 * \text{Time}) + 0.5$$
$$\text{SOA Complexity} = 0.5 / (0.1 * \text{Time} + 0.6)$$
$$\text{System Performance Requirement} = (90 * \text{Industry Performance Requirement}) / (40 + 100 * \text{Industry Performance Requirement}) + 0.1$$
$$\text{System Security Requirement} = (100 * \text{Industry Security Requirement}) / (40 + 100 * \text{Industry Security Requirement}) + 0.2$$

Constants

Availability of Services = **0.5**
Delay with SOA = **9** (assuming positive IT–Business Discrepancy)
Delay with Traditional Technology = **3** (assuming positive IT–Business Discrepancy)
Implementation Effectiveness = **0.7**
Leader’s Attitude toward Change = **0.5**
Number of Alternative Technologies = **3**
Relative Advantage of SOA = **0.6**
SOA Cost-Effectiveness = **0.5**
Standardization of Technology = **0.6**

Initial Values

Adopters(0) = Total Population * Initial Adopter Fraction
Initial Adopter Fraction = **0.1**
IT–Business Discrepancy(0) = **0**
Potential Adopters(0) = Total Population * (1 – Initial Adopter Fraction)
Total Population = 100 (as a percentage)

Note: Items in boldface represent organization- and industry-specific parameters.